

DAY 1

J

JAVA PROGRAMMING — FOUNDATIONS

Introduction · Print Statements · Data Types · Operators

8 TOPICS COVERED

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1. About Java & Program Structure

2. Types of Print Statements

3. Print Your Name

4. Print Name, Department, Age

5. Data Types in Java

6. Student Details Using Data Types

7. Operators in Java

8. Arithmetic Operations (+, -, ×, ÷)

Java is an object-oriented language. Every Java program lives inside a class, and the program always starts running from the main method.

C

class

A container that holds your program's code.

M

public static void main

The entry point — Java starts executing here.

S

System.out

Used to print output to the console.

PROGRAM SKELETON

```
public class Main {  
    public static void main(String[] args) {  
  
        // your code goes here  
  
    }  
}
```



Types of Print Statements



Java gives three ways to print output — each behaves a little differently.

1 `println()`

Prints the value, then moves to a new line.

2 `print()`

Prints the value, but stays on the same line.

3 `printf()`

Prints formatted output using format specifiers like `%d`, `%s`.

CODE

```
System.out.println("Hello");  
System.out.print("World");  
System.out.printf("Value = %d", 10);
```

CONSOLE OUTPUT

```
Hello  
WorldValue = 10
```

1

Print Your Name

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INPUT FORMAT

No input required — the name is assigned directly inside the program as a String variable.

O

OUTPUT FORMAT

name (String)

The name to be printed to the console

SAMPLE OUTPUT

```
My Name is Rahul
```



Java Code – Print Your Name

FULL PROGRAM

```
public class PrintName {  
    public static void main(String[] args) {  
  
        String name = "Rahul";  
        System.out.println("My Name is " + name);  
  
    }  
}
```

2

Print Name, Department, Age

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INPUT FORMAT

No input required — name, department and age are assigned directly inside the program.

O

OUTPUT FORMAT

name (String)

Name of the student

dept (String)

Department of the student

age (int)

Age of the student

SAMPLE OUTPUT

Name: Rahul

Department: Computer Science

Age: 20



Java Code – Name, Department, Age

FULL PROGRAM

```
public class StudentInfo {  
    public static void main(String[] args) {  
  
        String name = "Rahul";  
        String dept = "Computer Science";  
        int age = 20;  
  
        System.out.println("Name: " + name);  
        System.out.println("Department: " + dept);  
        System.out.println("Age: " + age);  
  
    }  
}
```

Java is strongly typed — every variable must be declared with a specific data type.

C

COMMON DATA TYPES

int (*age = 20*)

Stores whole numbers (no decimal point).

double (*cgpa = 8.7*)

Stores decimal / floating-point numbers.

String (*name = "Rahul"*)

Stores text — a sequence of characters.

char (*grade = 'A'*)

Stores a single character.

boolean (*pass = true*)

Stores only true or false.

I INPUT FORMAT**name** (*String*)

Name of the student (e.g. Rahul)

mark10, mark12, cgpa (*double*)

10th %, 12th % and CGPA (e.g. 92.5, 88.6, 8.7)

O OUTPUT FORMAT**name** (*String*)

Printed as-is

mark10, mark12, cgpa (*double*)

Printed as decimal values

SAMPLE OUTPUT

Name: Rahul

10th Mark %: 92.5

12th Mark %: 88.6

CGPA: 8.7



Java Code – Student Details



FULL PROGRAM

```
public class StudentDataTypes {  
    public static void main(String[] args) {  
  
        String name = "Rahul";  
        double mark10 = 92.5;  
        double mark12 = 88.6;  
        double cgpa = 8.7;  
  
        System.out.println("Name: " + name);  
        System.out.println("10th Mark %: " + mark10);  
        System.out.println("12th Mark %: " + mark12);  
        System.out.println("CGPA: " + cgpa);  
  
    }  
}
```

Arithmetic operators let you perform calculations on two or more values.

A

ARITHMETIC OPERATORS

+ $(a + b)$

Addition — adds two values.

- $(a - b)$

Subtraction — finds the difference between two values.

***** $(a * b)$

Multiplication — multiplies two values.

/ (a / b)

Division — divides one value by another.

% $(a \% b)$

Modulus — gives the remainder of a division.

I INPUT FORMAT**number_1** (*int*)

First number (e.g. 10)

number_2 (*int*)

Second number (e.g. 20)

O OUTPUT FORMAT**sum, difference** (*int*)

Results of addition and subtraction

product, quotient (*int / double*)

Results of multiplication and division

FORMULA $\text{sum} = n1 + n2$ $\text{difference} = n1 - n2$ $\text{product} = n1 * n2$ $\text{quotient} = n1 / n2$ **SAMPLE OUTPUT** $\text{number_1} = 10, \text{number_2} = 20$ $\text{Sum} = 30 \quad \text{Difference} = -10$ $\text{Product} = 200 \quad \text{Quotient} = 0.5$



Java Code – Arithmetic Operations

FULL PROGRAM

```
public class Operators {  
    public static void main(String[] args) {  
  
        int number_1 = 10;  
        int number_2 = 20;  
  
        int sum      = number_1 + number_2;  
        int difference = number_1 - number_2;  
        int product   = number_1 * number_2;  
        double quotient = (double) number_1 / number_2;  
  
        System.out.println("Sum = " + sum);  
        System.out.println("Difference = " + difference);  
        System.out.println("Product = " + product);  
        System.out.println("Quotient = " + quotient);  
  
    }  
}
```